

Paper 1 Section B

Solution	Marks	Remarks

2. In the marking scheme, answer marks of 'A' marks are awarded for a correct answer with a unit. If the answer should be in km, then cm and m are considered to be wrong units.

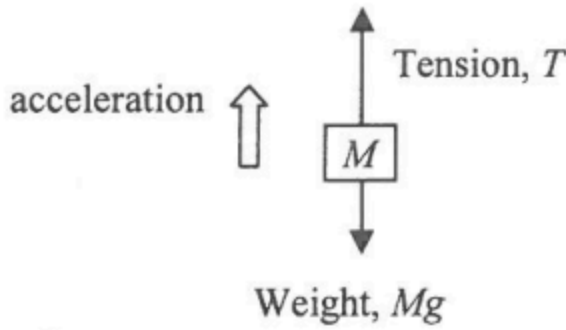
... parts each depending on the previous parts, method marks

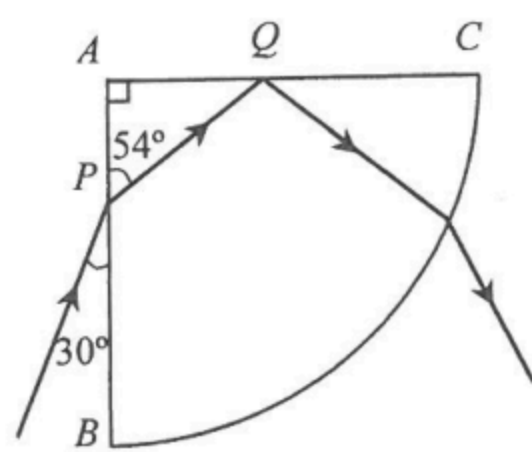
3. In a question consisting of several parts each depending on the previous part, or 'M' marks are awarded to steps/methods or substitutions correctly deduced from previous answers.
4. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

DSE Paper 1 Section A

Question No.	Key	Question No.	Key
1.	D (83)	26.	A (58)
2.	A (55)	27.	D (22)
3.	C (75)	28.	D (68)
4.	A (54)	29.	B (28)
5.	B (37)	30.	B (57)
6.	D (45)	31.	D (54)
7.	C (26)	32.	A (61)
8.	C (54)	33.	C (56)
9.	B (91)		
10.	B (66)		
11.	B (29)		
12.	B (58)		
13.	B (74)		
14.	A (80)		
15.	C (78)		
16.	A (76)		
17.	B (53)		
18.	C (41)		
19.	A (76)		
20.	C (44)		
21.	D (43)		
22.	D (61)		
23.	B (53)		
24.	C (20)		
25.	D (48)		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

4.	(a)	Arrows correct Labels (tension / T , weight / Mg)	1A 1A
		Or Either force correct and with label All correct.	1A 1A
			
		As M accelerates upward, $T - Mg = Ma$ i.e. $T = Mg + Ma$	1M
		Greater tension needed ($T > \text{original tension} = Mg$) and thus the spring extends further. i.e. M is closer to the bottom, smaller h in Figure 4.2	1A
			4
	(b)	Change in tension / reading = $2 \text{ N cm}^{-1} \times 0.5 \text{ cm} = 1 \text{ N}$ $\therefore T - Mg = 1 \text{ N}$ i.e. $T = 6 \text{ N}$	1A
		$T - Mg = Ma$	
		$1 \text{ N} = \frac{5 \text{ N}}{g} a$ (accept $M = 0.5 \text{ kg}$)	1M
		$a = \frac{1}{5}g$ or $0.2g$ (upward) (or $\frac{g}{5} = 2 \text{ m s}^{-2}$ or 1.96 m s^{-2})	1A
			3
	(c)	$a_Y = -0.5g = -g \cos \theta$ $\therefore \theta = 60^\circ > 45^\circ$, 'Landscape display' results.	1M 1A
		Or When the phone is rotated 45° clockwise, $a_Y = -g \cos 45^\circ$ or $-g \sin 45^\circ$ $= -\frac{g}{\sqrt{2}} = -0.71g$ or -6.94 m s^{-2} (or -7.07 m s^{-2}) 'Landscape display' results as $a_Y = -0.5g > -0.71g$.	1M 1A
		Or 'Landscape display' results as magnitude of $a_Y < \text{that at } 45^\circ$	1A
			2

Solution	Marks	Remarks
5. (a) $n = \frac{\sin i}{\sin r}$ (or $n_1 \sin \theta_1 = n_2 \sin \theta_2$) $= \frac{\sin(90^\circ - 30^\circ)}{\sin(90^\circ - 54^\circ)} = \frac{\sin 60^\circ}{\sin 36^\circ}$ $= 1.47$	1M 1A	Accept 1.47 to 1.50
	2	
(b) $\sin c = \frac{1}{n} = \frac{1}{1.47}$ $c = 42.7^\circ$ (for $n = 1.50$, $c = 41.8^\circ$) Because incident angle at face AC ($= 54^\circ$) $> c$ ($= 42.7^\circ$).	1M 1M	
	2	
(c) 		
Reflected ray correct, $i = r$ Emergent ray away from normal.	1A 1A	
	2	
(d) A spectrum is seen. Or splitting into different colours occurs.	1A 1A	
	1	

Solution	Marks	Remarks

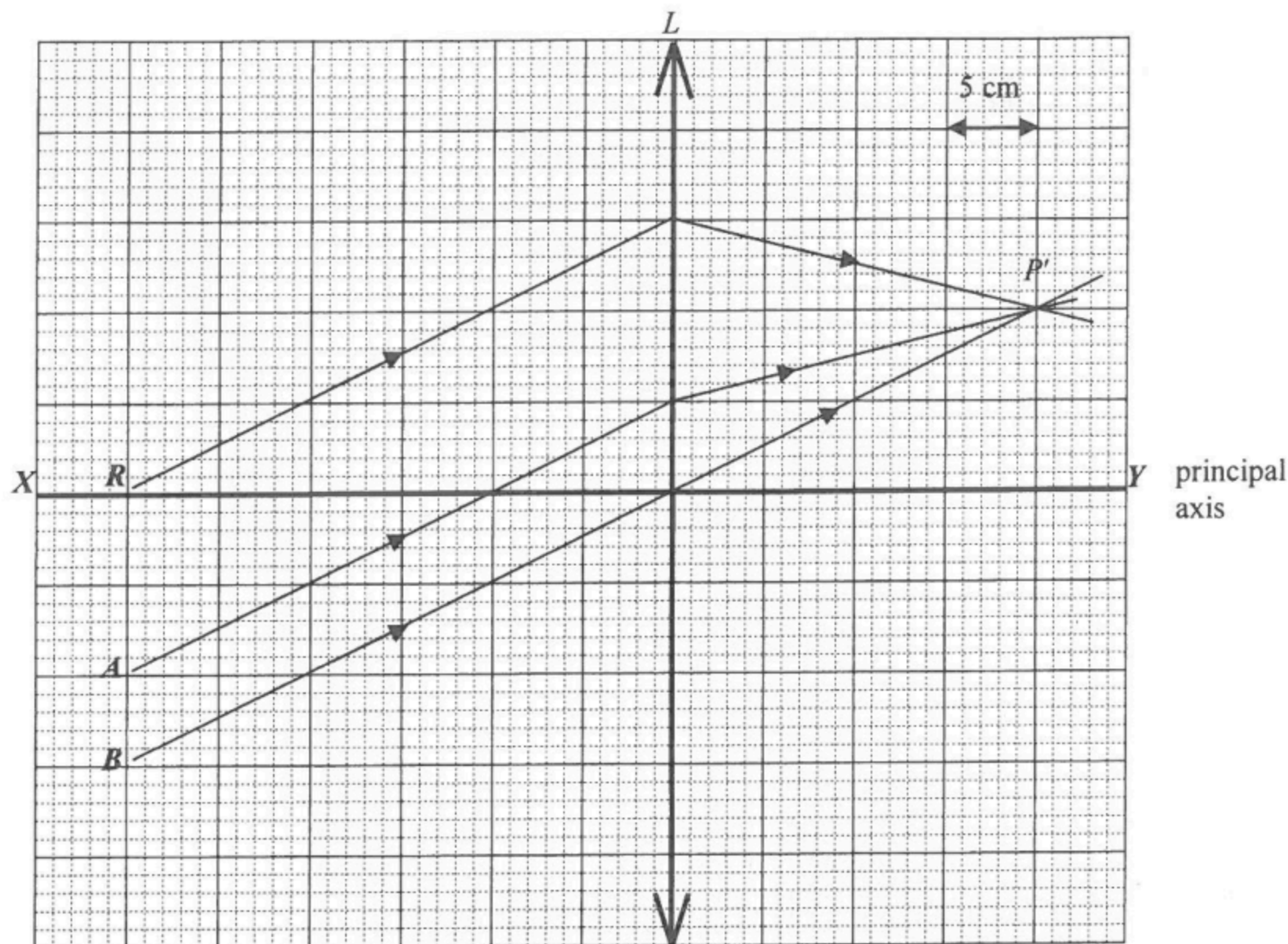
1.	(a)	5 min. (or 300 s)	1A
			1
	(b)	When the heater is switched off, its temperature is higher than the metal.	1A
		Or Heat continues to conduct to the metal until the same temperature is attained.	1A
		Or It takes time for them to attain the same temperature.	1A
			1
	(c)	(i)	1M
		$mc\Delta T = IVt$	
		$(0.80) c (45 - 20) = (4.0)(12)(5 \times 60)$	1A
		$c = 720 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$	2
		(ii)	1A
		Experimental result is higher than the actual value.	1A
		Not all the energy supplied by the heater goes to the metal.	
		Or Some energy is absorbed by the heater itself and / or by the thermometer.	1A
		Or Heat loss to the surroundings.	1A
			2
	(d)	Glass is not a good conductor of heat.	1A
		Or Prolonged time is required for the whole glass block to attain uniform temperature.	1A
		Or No insulating material is perfect, so heat loss is appreciable.	1A
			1

Solution	Marks	Remarks
2. (a) (i) V_0 = total volume of air required at P_0 , including the residual air in the ball $P_1 V_1 = P_0 V_0$ $(156 \text{ kPa}) (6000 \text{ cm}^3) = (100 \text{ kPa}) (V_0)$ $V_0 = 9360 \text{ cm}^3$ $\therefore$ volume of air = V_0 - volume of the basketball $= 9360 \text{ cm}^3 - 6000 \text{ cm}^3$ $= 3360 \text{ cm}^3$	1M 1A 1M	
Alternatively: $\therefore n = \frac{PV}{RT}$ $\frac{156V}{RT} - \frac{100V}{RT} = n'$ $\therefore V' = \frac{56V}{RT} \cdot \frac{RT}{100} = 3360 \text{ cm}^3$	1M 1M + 1A	
	3	
(ii) Number of strokes required = $3360 \text{ cm}^3 \div 120 \text{ cm}^3$ $= 28$	1A	
	1	
(b) Using kinetic theory, the pressure p is given by $pV = \frac{Nmc^2}{3} = \frac{2N}{3} \cdot \frac{mc^2}{2}$ As the volume and the temperature ($\propto$ kinetic energy of the air molecules) remain unchanged, the increase in pressure is due to the increase of number of air molecules hitting the wall of the container per unit time.	1A 1A	
Or Frequency of collision with the wall of container increases.	1A	
	2	

Solution	Marks	Remarks
3. (a) 0 – 10 s: constant / uniform acceleration	1A	

6.	(a)	Convex lens / Converging lens. Refracted ray of A after passing through L bends towards the principal axis (or optical centre) / converges / bends inward / bends downward.	1A 1A
		Or A real / an inverted image can be formed.	1A
		Or The object and the image are on opposite sides of the lens.	1A
			2

(b) (i)



Rays A and B correctly completed.
 P' correctly located.

(ii) $f = 20$ cm

(c) Ray R correctly completed.

(d) Use a screen to capture a/the (sharp) image (of a distant object).
The distance between the screen and the lens is f .

1M	
1M	
2	
1A	Accept 19 - 21 cm
1	
1M	
1	
1A	
1A	
2	

Solution	Marks	Remarks

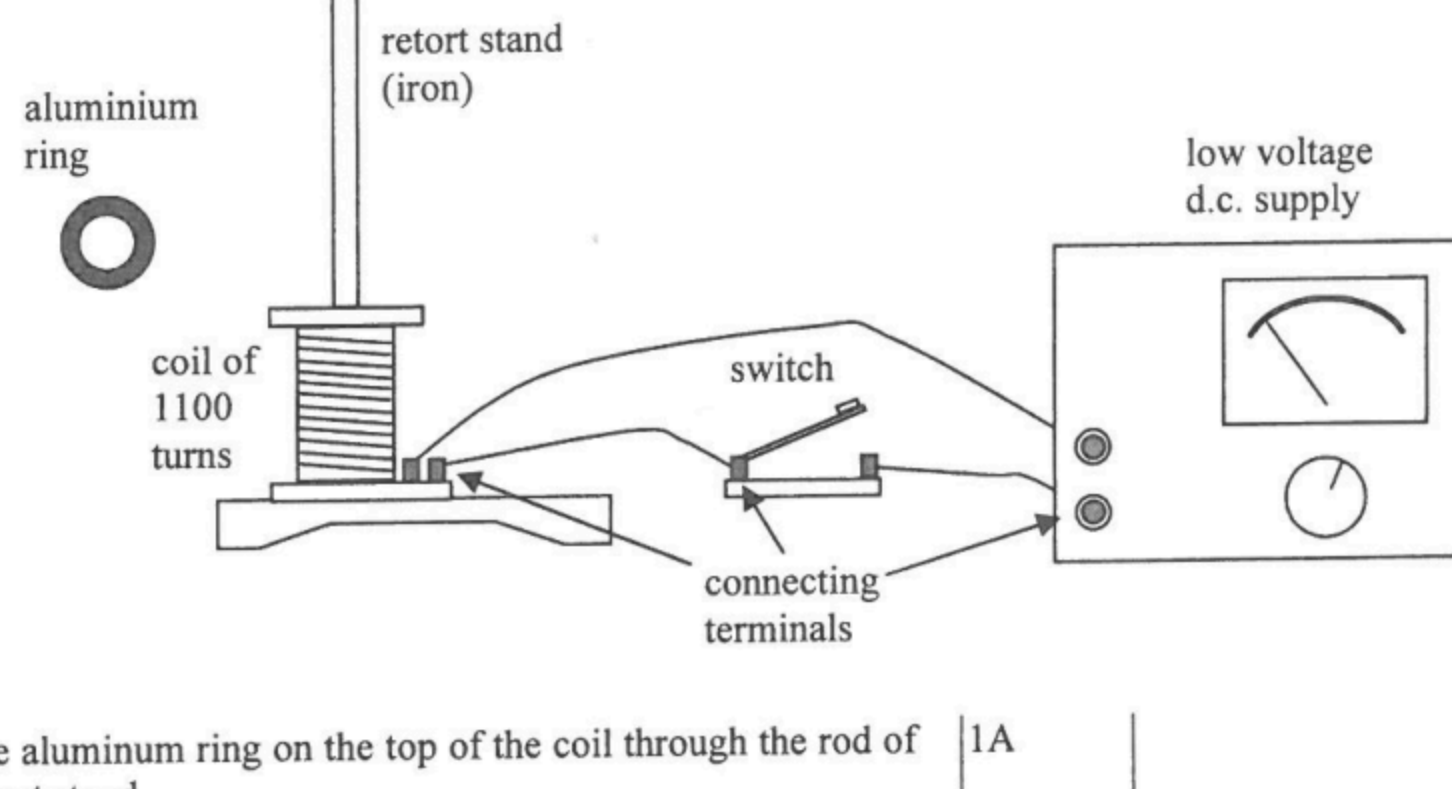
7. (a)	(i)	$\tan \theta = 0.38$ $\theta = 20.8^\circ$	1A	Accept 20.8° to 21°
			1	
	(ii)	$d \sin \theta = n\lambda$ As $d = (\frac{1}{300} \times 10^{-3})$, $(\frac{1}{300} \times 10^{-3}) \times \sin 20.8^\circ = 2\lambda$ $\lambda = 5.92 \times 10^{-7} \text{ m (or 592 nm)}$	1M	Accept $5.90 \times 10^{-7} \text{ m}$ to $5.97 \times 10^{-7} \text{ m}$
			1M	
			1A	
			3	
	(iii)	Smaller percentage error in x / the diffraction angle θ . Or Larger x, percentage error decreases.	1A	1
			1A	
	(b)	Repeat the procedures with the pin on the left-hand side / the other side of the observer. Take the average value of x obtained from both sides to calculate λ .		
1A				
1A				

Solution	Marks	Remarks
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8. (a)	$P = \frac{V^2}{R}$ $500 = \frac{220^2}{R}$ $R = 96.8 \, \Omega$	1A	Accept 97 Ω
		1	
(b)	As the total resistance of the circuit for mode X is doubled, total power dissipation = $\frac{V^2}{2R}$ $= \frac{220^2}{2 \times 96.8} = 250 \, \text{W}$	1M	
		1A	Accept 249 W for 97 Ω
	<i>Alternative method:</i> As the voltage across each heating element (1 and 2) is halved, power dissipated in each of the heating element $P_1 \text{ or } P_2 = \frac{500}{4} = 125 \, \text{W} \text{ (since } P \propto V^2 \text{)}$ Or $P_1 \text{ or } P_2 = \frac{V^2}{R} = \frac{110^2}{96.8} = 125 \, \text{W}$ Or $i = \frac{V}{R_1 + R_2} = \frac{220}{2 \times 96.8} = 1.14 \, \text{A}$ $P_1 \text{ or } P_2 = i^2 R = 1.14^2 \times 96.8 = 125 \, \text{W}$ Total power dissipation = $2 \times 125 \, \text{W} = 250 \, \text{W}$	1M 1M 1M 1A	
		2	
(c)	In mode Z, the equivalent resistance of the heating elements is the least as they are connected in parallel, hence, under the same voltage, the total power dissipation is the largest since $P = \frac{V^2}{R}$.	1A 1A	
		2	
(d) (i)	For mode Z, the total power dissipated = $500 + 500 = 1000 \, \text{W}$ $I_z = \frac{P}{V} = \frac{1000}{220} = 4.55 \, \text{A}$	1M+1M	
	<i>Alternative method:</i> $R_{eq} = \frac{96.8}{2} \, \Omega = 48.4 \, \Omega$ $I = \frac{220 \, \text{V}}{48.4 \, \Omega} = 4.55 \, \text{A}$ Most suitable value of fuse = 5 A	1M+1M 1A	
		3	
(ii)	Although the heater still works in either connection, it is dangerous for switch S to be fitted in wire B (neutral) as the heater / cable would still be live even when the switch was turned off.	1A 1A	1+1
		2	
(iii)	Wire C (Earth). Current would be conducted from the case through this wire to the Earth.	1A 1A	
		2	

Solution	Marks	Remarks

9. (a) Connect the coil to the terminals of the d.c. supply via the switch (diagram).



Put the aluminum ring on the top of the coil through the rod of the retort stand.
When closing the switch, the ring would shoot up the rod once, as the aluminum ring experiences a changing magnetic field produced by the coil at the start.
According to Lenz's Law, eddy currents flow in the ring to oppose the change.
Or A magnetic field in the opposite direction generated by the eddy currents repels the magnetic field produced by the coil.

However, when the current and thus the resulting magnetic field are constant, the ring would fall back to the coil as eddy currents no longer flow.

- (b) (i) The aluminum ring would float in the air.
(ii) The aluminum ring with a slit would remain stationary.

1A
1A
1A
1A
1A
1A
6
1A
1
1A
1

Solution	Marks	Remarks
10. (a) Alpha particles emitted can be stopped by the (thin) metallic casing. <u>Or</u> Shorter range / Lower penetrating power.	1A 1A	
(b) (i) $k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{87.74 \times 3.16 \times 10^7}$ $= 2.5 \times 10^{-10} \text{ s}^{-1} \text{ or } 7.9 \times 10^{-3} \text{ year}^{-1}$ Activity $A = kN$ $= \frac{\ln 2}{87.74 \times 3.16 \times 10^7} (3.2 \times 10^{25})$ $= 8.000 \times 10^{15} \text{ (Bq)}$	1M 1M 1A	
(ii) Power = Energy per decay $\times$ Activity $= 5.5 \text{ MeV} \times 8.000 \times 10^{15} \text{ Bq}$ $= 5.5 \times 10^6 \times 1.60 \times 10^{-19} \times 8.000 \times 10^{15}$ $= 7040 \text{ W or } 7.040 \text{ (kW)}$	1M 1A	
(iii) Power $\propto$ Activity Activity $\propto N$ $\therefore$ Percentage of power left $= \left(\frac{1}{2}\right)^{1/2} \times 100\%$ $= \left(\frac{1}{2}\right)^{36/87.74} \times 100\%$ $= 75.25\% \approx 75\%$ <u>Or</u> Percentage / fraction of power left = 3/4	1M 1A	
<u>Alternatively:</u> $N = N_0 e^{-kt}$ $\therefore$ Percentage of power left $= e^{-kt} \times 100\%$ $= e^{-(\ln 2 + 87.74) \times 36} \times 100\%$ $= e^{-0.2844} \times 100\%$ $= 75.25\% \approx 75\%$	1M 1A	

Paper 2

Section A : Astronomy and Space Science

1. D(55%)	2. A(31%)	3. C(53%)	4. A(39%)
5. B(58%)	6. D(50%)	7. C(30%)	8. B(36%)

Solution	Marks	Remarks
1. (a) (i) Luminosity of the star, $L = 4\pi R^2 \sigma T_s^4$ Power per unit area at distance d from the star $= \frac{L}{4\pi d^2} = \frac{R^2}{d^2} \sigma T_s^4$ Power absorbed $= \pi r^2 \times \frac{R^2}{d^2} \sigma T_s^4$	1M 1M	
(ii) At equilibrium, power absorbed = power radiated $\frac{R^2}{d^2} \pi r^2 \sigma T_s^4 = 4\pi r^2 \sigma T_p^4$ $\frac{R^2}{d^2} T_s^4 = 4 T_p^4$ $T_p^4 = \frac{R^2}{4d^2} T_s^4$ $T_p = \sqrt{\frac{R}{2d}} T_s$	1M 1M	1+1
(b) (i) $T_p = \sqrt{\frac{R}{2d}} T_s$ $= \sqrt{\frac{6.82 \times 10^8}{2 \times (0.84 \times 1.50 \times 10^{11})}} 5518$ $= 287 \text{ K (or } 14^\circ \text{C)}$	1M 1A	
(ii) The temperature is between 273 K and 373 K, (liquid) water is likely to exist on the planet. Hence the condition is favourable for life to exist.	1A 1A	
(iii) The equilibrium surface temperature is lower / decreases. A class K star is a cooler star than a class G star.	1A 1A	

Section B : Atomic World

1. A(71%)	2. D(45%)	3. C(57%)	4. C(49%)
5. D(52%)	6. B(46%)	7. A(43%)	8. B(60%)

Solution	Marks	Remarks
2. (a) A - (4): anode B - (3): condensing magnetic lens C - (1): objective magnetic lens D - (2): projection magnetic lens	2A	
(b) (i) K.E. = energy gain of the electron $\frac{1}{2}mv^2 = eV$ $(mv)^2 = 2meV$ $p = mv = \sqrt{2meV}$ $\therefore \lambda = \frac{h}{\sqrt{2meV}}$	1M 1M	
(ii) $\lambda = \frac{h}{\sqrt{2meV}}$ $= \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(10 \times 10^3)}}$ $\lambda = 1.2279 \times 10^{-11} \text{ m (or } 0.012 \text{ nm)}$	1M 1A	$1.20 \times 10^{-11} \sim 1.23 \times 10^{-11} \text{ m}$
(iii) Since the wavelength of the electron beam ($\sim 10^{-11} \text{ m}$) is smaller / shorter than that of visible light ($\sim 10^{-7} \text{ m}$), resolving power of a microscope, $\theta = \frac{1.22\lambda}{d}$, is greater with shorter wavelength (or less diffraction with shorter wavelength).	1A 1A	
(c) TEM. STM only reveals surface structure of specimen.	1A 1A	

Section C : Energy and Use of Energy

1. A(53%)	2. C(72%)	3. D(47%)	4. C(61%)
5. B(44%)	6. D(27%)	7. B(53%)	8. A(39%)

Solution	Marks	Remarks
3. (a) $E = VIt$ $23 \times 1000 = 220 \times 13 \times t$ $t = 8.04 \text{ (hours)}$	1M 1A	28951 s or 482.5 minutes
(b) (i) Converting electrical / energy from battery to KE / mechanical energy / force to drive the car / accelerate the car <u>or</u> Motor During braking, some of the kinetic energy of the wheels / vehicle is converted by the motor / generator / component X to electrical energy. The electrical energy is then stored in / used to charge the rechargeable battery.	1A 1A 1A	
(ii) High speed. When braking at high speed, the amount of kinetic energy that can be converted to electrical energy (to recharge the battery) is larger.	1A 1A	KE of the car = E ; Energy dissipated as heat = aE Energy to recharge battery = $(1 - a)E$ Fixed a . High speed $\Rightarrow$ More energy $(1 - a)E$ to recharge battery.
(iii) The mechanical braking system may come into play when the regenerative braking system fails. <u>Or</u> The mechanical braking system can lock the vehicle in position. <u>Or</u> The regenerative braking system will not work when the battery is fully charged.	1A 1A 1A	Accept same effect for high speed and low speed only if explanation has the meaning $\frac{E - aE}{E} = (1 - a)$
(c) Mode 2 (overall efficiency) = $45\% \times 60\% = 27\% > 20\%$ or 21% of the other two modes) No. Mode 3 practically has little or no emission of air pollutants.	1A 1A	

Section D : Medical Physics

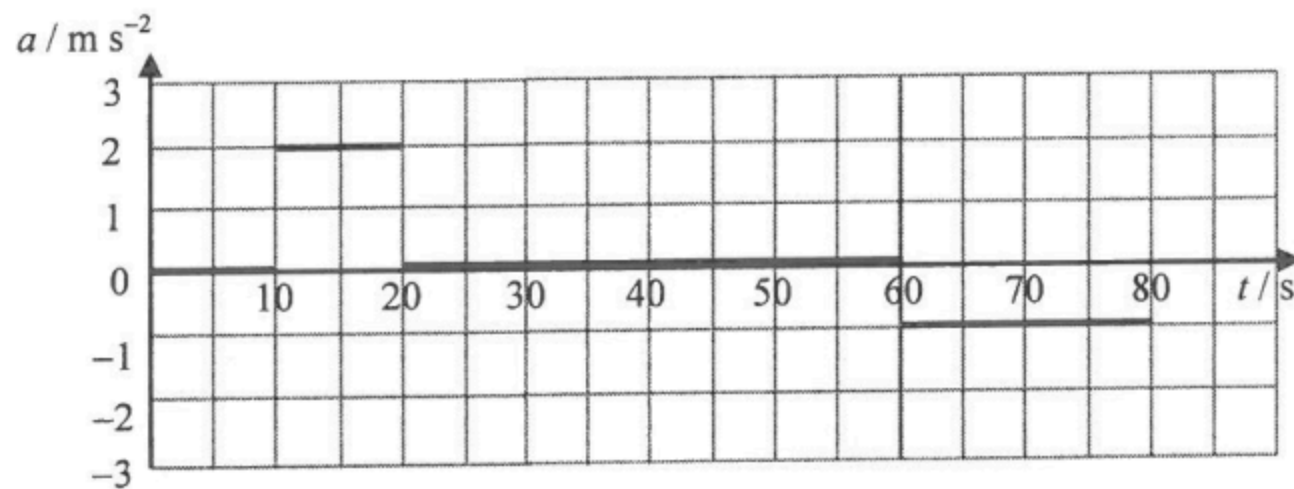
1. A(48%)	2. D(46%)	3. B(54%)	4. C(63%)
5. C(37%)	6. D(44%)	7. B(63%)	8. A(57%)

Solution	Marks	Remarks
4. (a) (i) When applying a potential difference across a small block of piezoelectric crystal inside the transducer, the crystal will be distorted; it will return to its original shape if the potential difference is removed, thus ultrasound waves will be generated due to its subsequent oscillations.	1A 1A	
(ii) Advantage : better resolution / clearer Disadvantage: greater attenuation / penetrate less	1A 1A	
(b) (i) $P = \frac{1}{u} + \frac{1}{v}$ or $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ $59 = \frac{1}{u} + \frac{1}{v}$ $v = 0.01695 \text{ m (or } 1.695 \text{ cm)} \approx 17 \text{ mm}$	1M 1A	Accept 16.9 ~ 17.0 mm
(ii) $\theta = \frac{1.22 \lambda}{d}$ $\theta = \frac{1.22 \times 5.35 \times 10^{-7}}{4.0 \times 10^{-3}}$ $= 1.63175 \times 10^{-4} \text{ (rad)} \approx 1.63 \times 10^{-4} \text{ (rad)}$	1M 1A	Accept 0.0093°
(iii) $\theta = \frac{r}{L}$ for small θ in radians $r = 1.632 \times 10^{-4} \times 0.30 \text{ m}$ $= 4.89525 \times 10^{-5} \text{ m (or } 0.0489525 \text{ mm)} \approx 49.0 \mu\text{m}$	1M 1A	Accept $\tan \theta = \frac{r}{L}$ Accept 48.9 ~ 49.0 μm

10 – 80 s: constant velocity / uniform motion

- (b) (i) Car B.
It is represented by the steepest slope between 10 s and 20 s for B.
 $a = \frac{20-0}{20-10} = 2 \text{ m s}^{-2}$

(ii)



Uniform acceleration between 10 s and 20 s & uniform deceleration between 60 s and 80 s.

Or Either acceleration or deceleration correct.

All correct.

- (c) (i) Total area of A by 20 s = $\frac{(10+20)}{2} \times 15 = 225 \text{ m}$
Total area of B by 20 s = $\frac{(10 \times 20)}{2} = 100 \text{ m}$
Separation of A & B, $s_{AB} = 225 \text{ m} - 100 \text{ m} = 125 \text{ m}$

- (ii) Let B meet A at time $(20 + T) \text{ s}$,
 $(v_B - v_A) \times T = s_{AB}$
 $(20 \text{ m s}^{-1} - 15 \text{ m s}^{-1}) \times T = 125 \text{ m}$
 $T = 25 \text{ s}$
 $\therefore B \text{ catches up with A at } t = (20 + 25) \text{ s} = 45 \text{ s}.$

Or $s_A + v_A \times T = s_B + v_B \times T$
 $225 \text{ m} + 15 \text{ m s}^{-1} \times T = 100 \text{ m} + 20 \text{ m s}^{-1} \times T$
Or Accept using graphical method and read the answer from the graph

- (d) Let F_A and F_B be the driving forces developed by the engines of A and B respectively.
Ratio of power developed by cars A and B,
 $P_A : P_B = F_A \times v_A : F_B \times v_B$
 $= v_A^2 \times v_A : v_B^2 \times v_B$
 $= 3^3 : 4^3 = 27 : 64$

Solution	Marks	Remarks